



SIMATS ENGINEERING

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Course Code: MLA0101

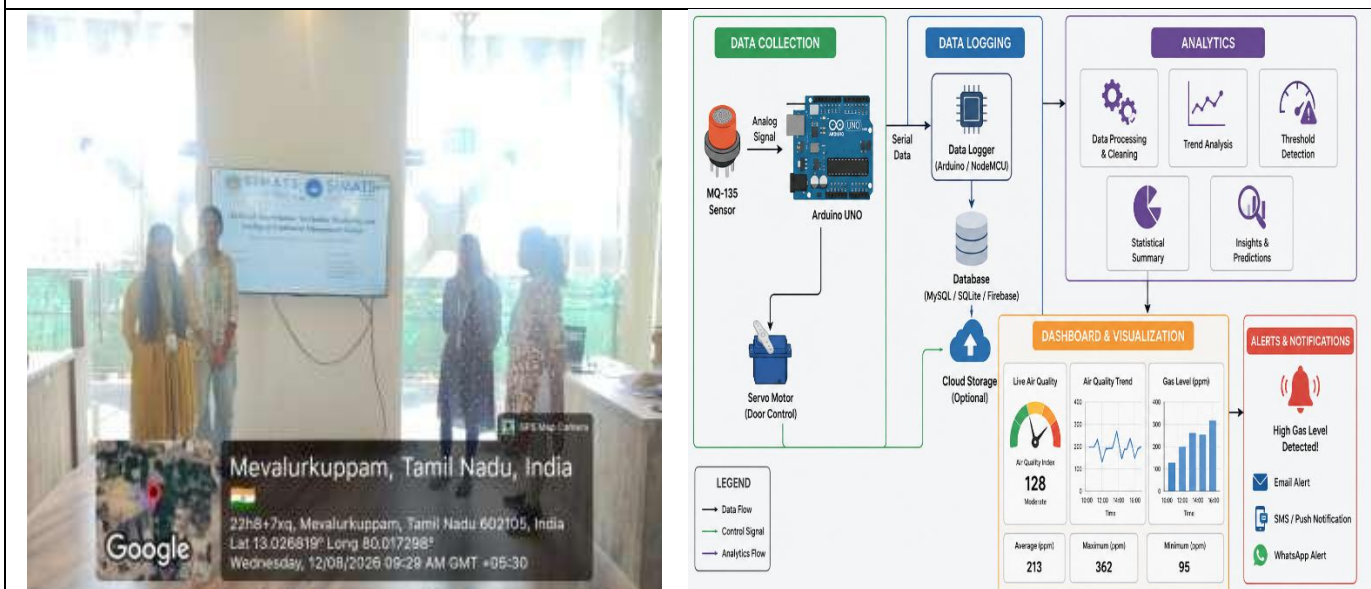
Slot: A

Course Name: Artificial Intelligence and Expert System

Course Faculty: Dr P Subramanian

Project Title: AI-Based Smart Indoor Air Quality Monitoring and Intelligent Ventilation System

Module Photographs: Module 4-Data Logging & Analytics



Project Description: (here you write what you did in this project (contribution) including Model Description

Project Description

This project develops a **low-cost AI-based Smart Indoor Air Quality Monitoring and Intelligent Ventilation Management System**. The system uses an **MQ-135 sensor and Arduino UNO** to continuously monitor indoor air-quality changes. The collected sensor data is analyzed using an **AI/ML model** to determine whether the air condition requires ventilation. When poor air quality is detected, the Arduino activates a **servo motor to automatically open the cardboard door**, allowing fresh air to enter. The system also supports **data logging and analytics** for observing air-quality trends and improving ventilation decisions.

Model Description:

The proposed AI model uses **sensor readings as input** and classifies the indoor condition as **Normal or Poor Air Quality**. Based on the model's prediction, the Arduino controls the servo motor to **close or open the ventilation door automatically**.

Student Signature

Guide Signature